



ICT Academy of Kerala FSD Projects Software Requirements Specification

Ecommerce Website

1.1 Purpose

The purpose of the online marketplace is to provide a platform for users to buy and sell products or services in a secure and user-friendly manner.

1.2 Scope

The online marketplace will include features such as user registration and authentication, product listings and search, shopping cart and checkout, user profiles , and an admin panel for management.

Functional Requirements

2.1 User Registration and Authentication

The system shall allow users to register by providing necessary information, including username, email, and password.

The system shall authenticate users by verifying their credentials during login.

2.2 Product Listings and Search

Sellers shall be able to create product listings by providing details such as title, description, price, and images.

Buyers shall be able to search and browse product listings based on various criteria, including keywords, categories, or location.

2.3 Shopping Cart and Checkout

The system shall provide a shopping cart feature for users to add products they want to purchase.

Users shall be able to view and modify the contents of their shopping cart.

The system shall facilitate a secure checkout process where users can provide shipping and payment information.

2.4 User Profiles

Users shall have the ability to create and manage their profiles, including personal information, contact details, and profile picture.

2.5 Admin Panel

The system shall provide an admin panel to manage user accounts, product listings, Admins shall have the ability to suspend or ban users, remove listings. Basically all CRUD operation

Non-Functional Requirements

3.1 Performance

Product search and browsing should be fast and efficient, even with a large number of listings.

3.2 Security

User authentication and sensitive data (e.g., passwords, payment information) should be securely stored and transmitted.

The system should implement measures to prevent unauthorized access, data breaches, or DB injection attacks.

3.3 Usability

The user interface should be intuitive, visually appealing, and responsive across different devices and screen sizes.

The system should provide clear and helpful error messages to assist users in case of input errors or system failures.

Constraints

4.1 Technology Stack

The system will be developed using the MERN stack (MongoDB, Express.js, React, and Node.js).

The payment gateway integration will be done using a specified payment provider's API.

4.2 Browser Compatibility

The system should be compatible with major web browsers such as Chrome, Firefox, and Safari.

General

Users

1. Clients/E-commerce users
2. Admin (to control product listing and users)

Minimum Pages required

1. Home Page
2. Login page
3. Registration Page
4. Product Listing page
5. Order shop cart page
6. Payment Page
7. User details page